

3D Coordinates Extension — Build the Snow Golem (Intermediate)

Topic: 3D Snow Golem (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 6) · **Time:** about 40 minutes

Build a **standing snow golem** in 3D — a snow body, a carved-pumpkin head, coal buttons down the front, and two stick arms. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

- Stand on your **home spot** (red block) every run. Move your feet, move your golem.
- 🎨 Snow = **snow block**. Head = **carved pumpkin**. Buttons + eyes = **coal block**. Arms = a **brown block** (oak log).

1 · Predict 🧠

Read each set. Write a sentence about what you will see.

```
place snow block at (3, 1, 1)
place snow block at (3, 2, 1)
place snow block at (3, 3, 1)
```

Only y changes. Does this snow tower go across, up, or deeper? Which number changes?


```
place snow block at (3, 3, 1)
place snow block at (3, 3, 2)
place snow block at (3, 3, 3)
```

Only z changes. Does it go across, up, or deeper? Which number changes?


```
place snow block at (3, 4, 1)
place coal block at (3, 4, 1)
```

Same (x, y, z). Do you see snow, coal, or both? Why?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — A coal button belongs **on the front** of the body at (3, 4, 1). The code has only **two** numbers, so it lands flat on the ground.

```
place coal block at (3, 4)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — An eye should sit **up high on the front** of the head at (4, 7, 1). The last two numbers are **swapped**, so it lands in the wrong place.

```
place coal block at (4, 1, 7)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Walk around the 🧊 snow golem in the world. Then build a snow golem on your home spot.

Snow body: fill snow blocks from (2, 1, 1) to (4, 5, 1) — a tall tower.

Pumpkin head on top (bigger y): fill carved pumpkin from (2, 6, 1) to (4, 7, 1).

Two coal eyes on the head (front): coal block at (2, 7, 1) and (4, 7, 1).

Two coal buttons on the body (front): coal block at (3, 4, 1) and (3, 2, 1).

Two arms out the sides: a brown block at (1, 4, 1) and (5, 4, 1).

Add depth: build the snow body and pumpkin head again at $z = 2$. The eyes, buttons, and arms stay on the **front only** ($z = 1$).

💡 Tip: the head goes **up** (bigger y). The arms poke **out** ($x = 1$ and $x = 5$). A bigger z means **deeper** into the wall.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the golem. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your **one** video to teacher on KakaoTalk.